

HW 1 - Dot Product

$$\textcircled{1} \quad \vec{a} \cdot \vec{b} = 2(-5) + 2(5) + 3(2) = \boxed{6}$$

$$\textcircled{2} \quad \|\vec{u}\| = \sqrt{100 + 64 + 81} = \sqrt{245} \approx \boxed{15.65} \text{ or } 7\sqrt{5}$$

$$\|\vec{v}\| = \sqrt{9 + 64 + 25} = \sqrt{98} \approx \boxed{9.9} \text{ or } 7\sqrt{2}$$

$$\vec{u} \cdot \vec{v} = -10(-3) + (-8)(8) + 9(5) = \boxed{11}$$

$$\textcircled{3} \quad \|\vec{v}\| = \sqrt{49 + 1 + 100} = \sqrt{150}$$

$$\|\vec{w}\| = \sqrt{36 + 4 + 9} = \sqrt{49} = 7$$

$$\Rightarrow \vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$$

$$\vec{v} \cdot \vec{w} = 7(6) + 1(2) + 10(3) = 74 \Rightarrow \cos \theta = \frac{74}{7\sqrt{150}} \approx \boxed{0.86}$$

$$\begin{aligned} \textcircled{4} \quad 3\vec{u} \cdot (8\vec{u} - 9\vec{v}) &= 24\vec{u} \cdot \vec{u} - 27\vec{u} \cdot \vec{v} = 24\|\vec{u}\|^2 - 27\vec{u} \cdot \vec{v} \\ &= 24(1) - 27(9) = \boxed{-219} \end{aligned}$$

$$\textcircled{5} \quad \vec{F} = [-2, -3, 3]$$

$$\vec{v} = [-4+5, -3, -4] = [1, -3, -4]$$

$$W = -2(1) + (-3)(-3) + 3(-4) = \boxed{-5}$$

$$\textcircled{6} \quad \vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta = (4)(7) \cos\left(\frac{\pi}{3}\right) = \boxed{14}$$

$$\textcircled{7} \quad -34b + b^3 + 9b = b^3 - 25b = b(b-5)(b+5)$$

$$\boxed{b = 0, \pm 5}$$

$\textcircled{8}$

(a) NEG

(b) POS

(c) POS

(d) ZERO

(e) POS

(f) NEG

(g) ZERO

(h) NEG